WEEK 1 · LESSON 5 OF 24 · HANDOUT

Files, Modules and Libraries

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=eWRfhZUzrAc — open the lesson online to watch inside the course.

Your goal in this lesson: Read and write text and CSV files, use Python's built-in modules, and import the data libraries you will use for the rest of the course.

Saving and reading a text file

```
with open("notes.txt", "w") as f:
    f.write("Week 1 notes\n")
    f.write("Python is readable.\n")

with open("notes.txt") as f:
    text = f.read()
print(text)
```

- `"w"` means **write** (it replaces the file). `"a"` means **append** (add to the end). No letter means **read**.
- `with` closes the file for you when the block ends — always use it.
- In Colab, files appear in the **folder icon** on the left. They disappear when the session ends unless you download them or save to Google Drive.

CSV files: tables in plain text

A **CSV** (comma-separated values) file is the most common way to share data. Each line is a row; commas separate the columns:

```
product,price,quantity
Rice 5kg,7500,3
Milk 1L,800,10
```

Write and read one with the built-in `csv` module:

```
import csv

rows = [
    ["product", "price", "quantity"],
    ["Rice 5kg", 7500, 3],
    ["Milk 1L", 800, 10],
    ["Sugar 1kg", 1600, 4],
]
with open("sales.csv", "w", newline="") as f:
    csv.writer(f).writerows(rows)

total = 0
with open("sales.csv") as f:
    for row in csv.DictReader(f):
        total += int(row["price"]) * int(row["quantity"])
print(f"Total sales: {total:,} RWF")
```

Notice `int(...)`: everything read from a CSV arrives as **text**. Next week, Pandas will read CSV files in one line and convert types for you.

Modules: code other people wrote

A **module** is a file of ready-made Python code. Python includes many:

```
import math
import random
import statistics

print(math.sqrt(144))                      # 12.0
print(round(math.pi, 4))                   # 3.1416
print(statistics.mean([72, 45, 88, 61]))   # 66.5
print(statistics.median([72, 45, 88, 61]))  # 66.5

random.seed(1)                    # same "random" numbers every run
print(random.randint(1, 6))       # roll a die
print(random.choice(["MoMo", "Cash", "Card"]))
```

`random.seed()` makes random results **repeatable**. AI researchers always set a seed so others can reproduce their results.

Libraries: the AI toolkit

Bigger collections of modules are called **libraries** (or packages). These are the ones you will use:

Library	Short name	What it does	Week
NumPy	`np`	Fast maths on arrays of numbers	2–4
Pandas	`pd`	Tables of data (like Excel in code)	2, 4
Matplotlib	`plt`	Charts and graphs	2–4
scikit-learn	`sklearn`	Ready-made machine-learning models	Next course

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

print(np.__version__, pd.__version__)
```

The short names (`np`, `pd`, `plt`) are a convention — every tutorial uses them, so you should too.

On your own computer you install a library once with `pip install pandas` in a terminal. In Colab they are already installed; if one is missing, run `!pip install name` in a cell.

Key terms

- **File mode** — `"r"` read, `"w"` write (replace), `"a"` append.
- **CSV** — a text file where commas separate columns.
- **Module** — a file of reusable Python code (`math`, `random`, `csv`).
- **Library / package** — a larger collection of modules (NumPy, Pandas).
- **import** — load a module or library so you can use it.
- **Seed** — a starting value that makes random results repeatable.

Remember this: Use `with open(...)` for files and the `csv` module for simple tables. Import modules instead of writing everything yourself — and meet NumPy, Pandas and Matplotlib, your tools for the rest of the course.

Name: _____ Date: _____

Activity

Task 1

Write a CSV file `students.csv` with columns `name, district, mark` and 5 rows.

Task 2

Read it back and print the average mark and the name of the top student.

Task 3

Use `random` with `seed(7)` to choose 3 random students for a presentation.

Task 4

Import NumPy and Pandas and print their versions.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does the mode `"a"` do in `open("log.txt", "a")`?

   A. Reads the file
   B. Replaces the file
   C. Deletes the file
   D. Adds new text to the end of the file

2. Why use `with open(...) as f:`?

   A. It encrypts the file
   B. It makes the file bigger
   C. It closes the file automatically when the block ends
   D. It is required for printing

3. A value read from a CSV with the csv module is…

   A. A list
   B. Always an int
   C. Always a float
   D. Text (a string) until you convert it

4. What does `random.seed(1)` do?

   A. Plants a tree
   B. Makes the random results repeatable
   C. Stops randomness forever
   D. Deletes random numbers

5. Which library is used for tables of data, like Excel in code?

   A. math
   B. csv
   C. Matplotlib
   D. Pandas

6. What is the usual short name for NumPy?

   A. np
   B. nu
   C. py
   D. num

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